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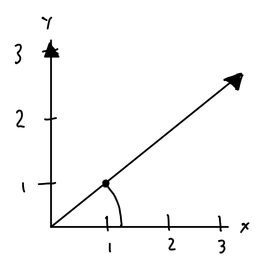
Signal Processing: HW2

① Inner Product

a) consider the vectors $\vec{v} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \vec{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

- draw them in the xy plane



formula:

$$\cos\theta = \frac{\langle \vec{v} | \vec{w} \rangle}{\|\vec{v}\| \|\vec{w}\|}$$

vectors:

$\vec{v} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \vec{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\vec{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\|\vec{w}\| = \sqrt{1^2 + 1^2}$

$\sqrt{2} = 1.414$

- inner product $= \langle \vec{v} | \vec{w} \rangle$

↳ $\boxed{v_x \times w_x + v_y \times w_y}$

$\boxed{0 \cdot 1 + 3 \cdot 1 = 3}$

$3 > 0 \rightarrow$ positively aligned

- angle

$$\cos\theta = \frac{3}{(3)(1.414)} = \frac{1}{1.414} = 0.707$$

$$\frac{3}{(3)(\sqrt{2})} \rightarrow \frac{1}{\sqrt{2}}$$

$$\cos\theta = \frac{1}{\sqrt{2}}$$

- Solve for the angle

$$\theta = \arccos\left(\frac{1}{\sqrt{2}}\right) = 45° \rightarrow \boxed{\frac{\pi}{4} \text{ radians}}$$

* angle between $\vec{v}$ & $\vec{w}$ = $45°$ ($\pi/4$ rad)

* inner product = 3

* lengths = $|v| = 3$, $|w| = \sqrt{2}$

b) projection of $\vec{w}$ onto $\vec{v}$

$$\vec{w_1} = \frac{\langle \vec{w_1} | \vec{w_2} \rangle}{\langle \vec{w_1} | \vec{w_1} \rangle}$$

given $\vec{v} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$, $\vec{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\hookrightarrow \|\vec{v}\| = 3$, $\|\vec{w}\| = \sqrt{2}$

* projection of $\vec{w}$ onto $\vec{v}$ =

$$\vec{v} = \frac{\langle \vec{v} \mid \vec{w} \rangle}{\langle \vec{v} \mid \vec{v} \rangle}$$

* projection of $\vec{v}$ onto $\vec{w}$ =

$$\vec{w} = \frac{\langle \vec{w} \mid \vec{v} \rangle}{\langle \vec{w} \mid \vec{w} \rangle}$$

* inner prods

$$\begin{bmatrix} 0 \\ 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\langle \vec{v} \mid \vec{w} \rangle = (0)(1) + (3)(1) = 3$$

$$\langle \vec{v} \mid \vec{v} \rangle = (0)(0) + (3)(3)$$

$$(0)^2 + (3)^2 = 9$$

$$\langle \vec{v} \mid \vec{w} \rangle = (1)^2 + (1)^2 = 2$$

* projection $\vec{w}$ onto $\vec{v}$

$$\vec{V} = \frac{\langle \vec{v} \mid \vec{w} \rangle}{\langle \vec{v} \mid \vec{v} \rangle} \rightarrow$$

$v_x = 0$
$v_y = 3$
$w_x = 1$
$w_y = 1$

$$\vec{v} \times \frac{3}{\langle \vec{v} \mid \vec{v} \rangle} \longrightarrow \vec{v} \times \frac{3}{9}$$

9

$$\begin{bmatrix} 0 \\ 3 \end{bmatrix} \cdot \frac{3}{9} = \begin{bmatrix} 0 \cdot \frac{3}{9} \\ 3 \cdot \frac{3}{9} \end{bmatrix} = \begin{bmatrix} 0 \\ \cancel{\frac{9}{9}}\ 1 \end{bmatrix}$$

$$\vec{w} \text{ onto } \vec{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

* length of projection vector

$$\| [x \ y]^T \| = \sqrt{x^2 + y^2}$$

$$\left\|\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\| = \sqrt{(0)^2 + (1)^2} =$$

$$\sqrt{1} = \boxed{1}$$

* projection of $\vec{v}$ onto $\vec{w}$

$$\vec{w} \cdot \frac{\langle \vec{w} | \vec{v} \rangle}{\langle \vec{w} | \vec{w} \rangle}$$

$\langle \vec{w} | \vec{v} \rangle$

$\hookrightarrow (w_x)(v_x) + (w_y)(v_y) =$

$(1)(0) + (1)(3) = 0 + 3 = 3$

$\langle \vec{w} | \vec{w} \rangle =$

$(1)(1) + (1)(1) = 2$

$\vec{w} \times \frac{3}{2}$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \frac{3}{2} = \begin{bmatrix} 1 \cdot \frac{3}{2} \\ 1 \cdot \frac{3}{2} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}$$

$$\left[\frac{3}{2}, \frac{3}{2}\right]^T$$

* length

$$\left\| \begin{bmatrix} \frac{3}{2} \\ \frac{3}{2} \end{bmatrix} \right\| = \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{3}{2}\right)^2} =$$

$$\sqrt{\frac{9}{4} + \frac{9}{4}} = \sqrt{\frac{\cancel{18}^{\,9}}{\cancel{4}_{\,2}}} = \boxed{\frac{3}{\sqrt{2}}}$$

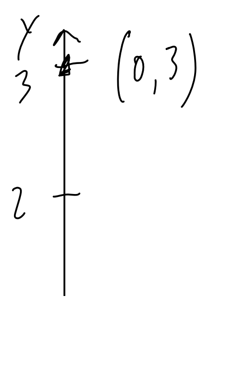
* length of original vector

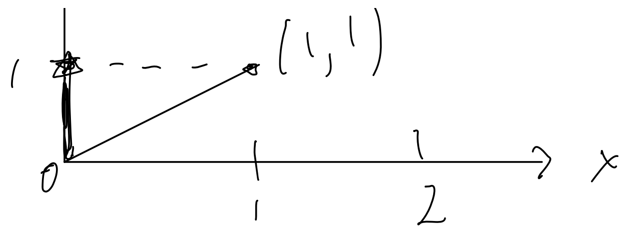
$$\|\vec{v}\| = \sqrt{(0)^2 + (3)^2} = \sqrt{0+9}$$

$$\boxed{3}$$

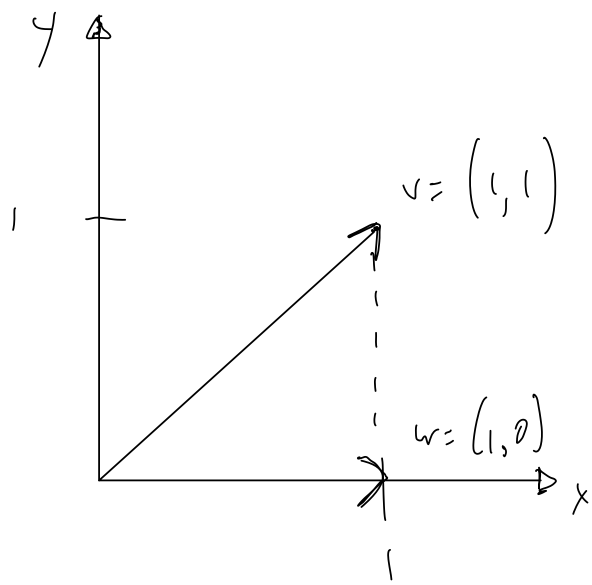
$$\|\vec{w}\| = \sqrt{(1)^2 + (1)^2} = \boxed{\sqrt{2}}$$

* projection of $\vec{w}$ onto $\vec{v}$





projection of $\vec{v}$ onto $\vec{w}$



c) inner products & relationship

$\vec{v} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$, $\vec{w} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

* for real vectors, the inner product is just the dot product

$\langle \vec{v} | \vec{w} \rangle = \vec{v}^T \vec{w} = \begin{bmatrix} 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0 \cdot 1 + 3 \cdot 1 = 3$

$\langle \vec{w} | \vec{v} \rangle = \vec{w}^T \vec{v} = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \end{bmatrix} = 1 \cdot 0 + 1 \cdot 3 = 3$

* symmetry check

$\langle \vec{v} | \vec{w} \rangle = \langle \vec{w} | \vec{v} \rangle = 3$

↳ they are equal (symmetric) for real vectors

d) vectors $\vec{z_1} = \begin{bmatrix} 0 \\ j \\ 1 \end{bmatrix}$ and $\vec{z_2} = \begin{bmatrix} 1 \\ j \\ j \end{bmatrix}$

$$\langle z_1 | z_2 \rangle = \vec{z}_1^{\,\dagger} \vec{z}_2$$

* $\vec{z_1} = \begin{bmatrix} 0 \\ j \\ 1 \end{bmatrix} \Rightarrow z_1^{\dagger} = \begin{bmatrix} 0^* & j^* & 1^* \end{bmatrix} = \begin{bmatrix} 0 & -j & 1 \end{bmatrix}$

$$\langle z_1 | z_2 \rangle = \begin{bmatrix} 0 & -j & j \end{bmatrix} = 0(1) + (-j)(j) + 1(j)$$

$$(-j)(j) = -j^2 = 1$$

$$\langle z_1 | z_2 \rangle = 0 + 1 + j = \boxed{1 + j}$$

* $\vec{z_2} = \begin{bmatrix} 1 \\ j \\ j \end{bmatrix} \Rightarrow z_2^{\dagger} = \begin{bmatrix} 1^* & j^* & j^* \end{bmatrix} = \begin{bmatrix} 1 & -j & -j \end{bmatrix}$

* $\langle z_2 | z_1 \rangle = \begin{bmatrix} 1 & -j & -j \end{bmatrix} \begin{bmatrix} 0 \\ j \\ 1 \end{bmatrix} =$

$1(0) + (-j)(j) + (-j)(1)$

$(-j)(j) = 1, \quad (-j)(1) = -j$

$\langle z_2 | z_1 \rangle = 0 + 1 - j = \boxed{1 - j}$

* conjugate symmetry

$(1+j)^* = 1 - j$

$$\boxed{\langle z_1 | z_2 \rangle = 1 + j, \quad \langle z_2 | z_1 \rangle = 1 - j}$$

2) Gram-Schmidt Orthogonalization

$\vec{w_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{w_2} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\vec{w_3} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

a) independent?

$c_1\vec{w_1} + c_2\vec{w_2} + c_3\vec{w_3} = \vec{0}$

$c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} =$

$\begin{bmatrix} c_1 + c_2 + c_3 \\ c_2 + c_3 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\begin{cases} c_1 + c_2 + c_3 = 0 \\ c_2 + c_3 = 0 \\ c_3 = 0 \end{cases} \rightarrow$

$c_3 = 0$

$c_2 + 0 = 0 \Rightarrow 0$

$c_1 + 0 + 0 = 0 \Rightarrow c_1 = 0$

* the only solution is $c_1 = c_2 = c_3 = 0$

↳ vectors are linearly independent

$$\boxed{\vec{w}_1, \vec{w}_2, \vec{w}_3 \text{ are linearly independent}}$$

b) Are the three $\vec{w}_k$ orthogonal?

$\vec{w}_1 \cdot \vec{w}_2 = 1 \neq 0$

$\vec{w}_1 \cdot \vec{w}_3 = 1 \neq 0$

$\vec{w}_2 \cdot \vec{w}_3 = 2 \neq 0$

$\vec{b_1} = \vec{w}_1,$

$$\vec{b_2} = \vec{w_2} - b_1 \frac{\langle \vec{b_1}, \vec{w_2} \rangle}{\langle \vec{b_1}, \vec{b_1} \rangle}$$

$$\vec{b_3} = \vec{w_3} - b_1 \frac{\langle \vec{b_1}, \vec{w_3} \rangle}{\langle \vec{b_1}, \vec{b_1} \rangle} - b_2 \frac{\langle \vec{b_2}, \vec{w_3} \rangle}{\langle \vec{b_2}, \vec{b_2} \rangle}$$

$$\vec{b_1} = \vec{w_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\langle \vec{b_1}, \vec{w_2} \rangle = \vec{w_1} \cdot \vec{w_2} = 1, \quad \langle \vec{b_1}, \vec{b_1} \rangle = \vec{w_1} \cdot \vec{w_1} = 1$$

$$\Rightarrow \vec{b_2} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} (1) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\langle \vec{b_1}, \vec{w_3} \rangle = \vec{w_1} \cdot \vec{w_3} = 1, \quad \langle \vec{b_1}, \vec{b_1} \rangle = 1;$$

$$\langle \vec{b_2}, \vec{w_3} \rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1, \quad \langle \vec{b_2}, \vec{b_2} \rangle = 1$$

$$\Rightarrow \vec{b_3} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} (1) - \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} (1) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

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* orthogonal basis

$$\vec{b_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{b_2} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \vec{b_3} : \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

3) Orthonormal Bases

three real vectors $\vec{v} = \begin{bmatrix} v_0 \\ v_1 \\ v_2 \end{bmatrix} = \hat{e}_0 v_0 + \hat{e}_0 v_1 + \hat{e}_0 v_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$

$\hat{e}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \hat{e}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$\vec{w_1} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \vec{w_2} = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}, \vec{w_3} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$

a) assume $c_1 \vec{w_1} + c_2 \vec{w_2} + c_3 \vec{w_3} = \vec{0}$

① $c_1 + 2c_2 + 0 \cdot c_3 = 0 \Rightarrow c_1 + 2c_2 = 0$

② $c_1 - c_2 + c_3 = 0$

③ $c_1 - c_2 - c_3 = 0$

$(2)(3)$: $(c_1 - c_2 + c_3) + (c_1 - c_2 - c_3) = 2(c_1 - c_2) = 0$

so $c_1 = c_2$

* $c_1 + 2c_2 = 0 \Rightarrow c_2 + 2c_2 = 3c_2 \Rightarrow c_2 = 0$

$c_1 = 0$ as well

* c_3 with $(2)(3)$

$0 - 0 + c_3 = 0 \Rightarrow c_3 = 0$

* $c_1 = c_2 = c_3 = 0$ works, so the vectors are

$\boxed{\text{linearly independent}}$

b) Orthogonal?

$w_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $w_2 = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}$, $w_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$

$w_1 \cdot w_2 = 1 \cdot 2 + 1 \cdot (-1) + 1 \cdot (-1) = 0$ ✓

$w_1 \cdot w_3 = 1 \cdot 0 + 1 \cdot 1 + 1 \cdot (-1) = 0$ ✓

$w_2 \cdot w_3 = 2 \cdot 0 + (-1) \cdot 1 + (-1) \cdot (-1) = 0$ ✓

$\boxed{\text{mutually orthogonal}}$

c) Norms

$\|\vec{w_1}\| = \sqrt{1^2+1^2+1^2} = \sqrt{3}$

$\|\vec{w_2}\| = \sqrt{2^2+(-1)^2+(-1)^2} = \sqrt{6}$

$\|\vec{w_3}\| = \sqrt{0^2+1^2+(-1)^2} = \sqrt{2}$

* orthonormal basis $\{\hat{b}_k\}$

$$b_1 = \frac{\vec{w}_1}{\|\vec{w}_1\|} = \frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\1\end{bmatrix}, \quad b_2 = \frac{\vec{w}_2}{\|\vec{w}_2\|} = \frac{1}{\sqrt{6}}\begin{bmatrix}2\\-1\\-1\end{bmatrix},$$

$$b_3 = \frac{\vec{w}_3}{\|\vec{w}_3\|} = \frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\-1\end{bmatrix}$$

d) Coordinate transformation matrix

$$T_{e\to b} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & \frac{-1}{\sqrt{6}} & \frac{-1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix}$$

* applied to $\vec{v}$

let $\vec{v} = \begin{bmatrix} v_0 \\ v_1 \\ v_2 \end{bmatrix}$

$$\boxed{\begin{aligned} v_1 &= \frac{1}{\sqrt{3}}(v_0 + v_1 + v_2), \\ v_2 &= \frac{1}{\sqrt{6}}(2v_0 - v_1 - v_2), \\ v_3 &= \frac{1}{\sqrt{2}}(v_1 - v_2) \end{aligned}}$$

4) Orthonormal Bases

4 vector $\vec{v} = \begin{bmatrix} v_0 \\ v_1 \\ v_2 \\ v_3 \end{bmatrix} = \hat{e}_0 v_0 + \hat{e}_1 v_1 + \hat{e}_2 v_2 + \hat{e}_3 v_3$

where $\hat{e_0} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, $\hat{e_1} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, $\hat{e_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, $\hat{e_3} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

$K = 0, 1, 2, 3$, and $n = 0, 1, 2, 3$

a) $\sum_{K=0}^{3} \vec{w} K$

$\vec{w}k$:

$$\vec{w}_0 = \begin{bmatrix} e^{j2\pi \cdot 0 \cdot 0/4} \\ e^{j2\pi \cdot 0 \cdot 1/4} \\ e^{j2\pi \cdot 0 \cdot 2/4} \\ e^{j2\pi \cdot 0 \cdot 3/4} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\vec{w}_2 = \begin{bmatrix} e^{j2\pi \cdot 2 \cdot 0/4} \\ e^{j2\pi \cdot 2 \cdot 1/4} \\ e^{j2\pi \cdot 2 \cdot 2/4} \\ e^{j2\pi \cdot 2 \cdot 3/4} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

$$\vec{w}_1 = \begin{bmatrix} e^{j2\pi \cdot 1 \cdot 0/4} \\ e^{j2\pi \cdot 1 \cdot 1/4} \\ e^{j2\pi \cdot 1 \cdot 2/4} \\ e^{j2\pi \cdot 1 \cdot 3/4} \end{bmatrix} = \begin{bmatrix} 1 \\ j \\ -1 \\ -j \end{bmatrix}$$

$$\vec{w}_3 = \begin{bmatrix} e^{j2\pi \cdot 3 \cdot 0/4} \\ e^{j2\pi \cdot 3 \cdot 1/4} \\ e^{j2\pi \cdot 3 \cdot 2/4} \end{bmatrix} = \begin{bmatrix} 1 \\ -j \\ -1 \end{bmatrix}$$

$-e^{j2\pi\cdot 3\cdot 3/4}-$ $\quad$ L J J

* coordinate-wise

- row $n=0: 1+1+1+1=4$
- row $n=1: 1+j+(-1)+(-j)=0$
- row $n=2: 1+(-1)+1+(-1)=0$
- row $n=3: 1+(-j)+(-1)+j=0$

$$\sum_{k=0}^{3} \vec{w}_k = \begin{bmatrix} 4 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

b) $w = e^{j2\pi/4} \in \{1, j, -1, -j\}$

each vector is $\vec{w}_k[n] = w^{kn}$

for $n = 0, 1, 2, 3$ & $k = 0, 1, 2, 3$

* linear combo $= 0$

$$c_0\vec{w}_0 + c_1\vec{w}_1 + c_2\vec{w}_2 + c_3\vec{w}_3 = \vec{0}$$

$(n=0)$ $c_0 + c_1 + c_2 + c_3 = 0$

$(n=1)$ $c_0 + jc_1 - c_2 - jc_3 = 0$

$(n=2)$ $c_0 - c_1 + c_2 - c_3 = 0$

$(n=3)$ $c_0 - jc_1 - c_2 + jc_3 = 0$

* case 1: $c_0 \neq 0$

→ adding $(n=0) + (n=2)$

$$(c_0 + c_1 + c_2 + c_3) + (c_0 - c_1 + c_2 - c_3) = 2c_0 + 2c_2 = 0 \Rightarrow c_2 = -c_0$$

→ adding $(n=1) + (n=3)$:

$$(c_0 + jc_1 - c_2 - jc_3) + (c_0 - jc_1 - c_2 + jc_3) = 2c_0 = 0 \Rightarrow c_0 = c_2$$

together

$c_2 = -c_0$ & $c_2 = c_0$ implies $c_0 = 0$

↳ contradicts $c_0 \neq 0$

* case 2: $c_0 = 0$

$c_1 + c_2 + c_3 = 0$

$j c_1 - c_2 - j c_3 = 0$

$-c_1 + c_2 - c_3 = 0$

$-j c_1 - c_2 + j c_3 = 0$

$\rightarrow$ adding $(n=0) + (n=2) = 0 \Rightarrow c_2 = 0$

$(n=0)$ gives $c_1 + c_3 = 0 \Rightarrow c_3 = -c_1$

$\rightarrow$ using $(n=1)$: $j(c_1 - c_3) = j(c_1 - (-c_1)) =$

$2 j c_1 = 0 \Rightarrow c_1 = 0 \Rightarrow c_3 = 0$

thus $c_0 = c_1 = c_2 = c_3 = 0$ is the only solution

$\boxed{\text{the 4 vectors are linearly independent}}$

c) 4 $\vec{w_k}$ orthogonal?

using $e^{j2\pi \frac{kn}{4}}$: $\left[\begin{matrix} k = 0, 1 \ldots 3 \\ n = 0, \ldots 3 \end{matrix}\right]$

↳ $e^{j\pi/2} = j$

$\vec{w_0} = [1, 1, 1, 1]^T$

$\vec{w_1} = [1, j, -1, -j]^T$

$\vec{w_2} = [1, -1, 1, -1]^T$

$\vec{w_3} = [1, -j, -1, j]^T$

* inner prod (in bra-ket)

complex vectors:

$$\langle \vec{w_k} | \vec{w_p} \rangle = \sum_{n=0}^{3} e^{j2\pi kn/4} \, e^{j2\pi pn/4} =$$

$$\sum_{n=0}^{3} e^{j2\pi (p-k) n/4}$$

* roots of unity

let $m = p - k$

$$\langle \vec{w}_k | \vec{w}_p \rangle = \sum_{n=0}^{3} \left(e^{i2\pi m/4} \right)^n$$

if $m = 0 \Rightarrow$ the sum is $1+1+1+1=4$

if $m \neq 0 \Rightarrow$ finite geometric series with ratio $w = e^{i2\pi m/4} \neq 1$

$$\langle \vec{w}_k | \vec{w}_p \rangle = \begin{cases} 4, & k = p \\ 0, & k \neq p \end{cases}$$

$\Rightarrow$ they're mutually orthogonal

$\langle v_k, v_k \rangle = N \Rightarrow$ orthonormal basis by dividing each by $\sqrt{N}$

* no need for Gram-Schmidt

d) Norms of orthogonal vectors

* normalize

$$\hat{\vec{b}}_k \triangleq \frac{1}{\|\vec{W}_k\|}\vec{W}_k = \frac{1}{2}\vec{W}_k, \quad k=0,1,2,3$$

written explicitly

- $\hat{\vec{b}}_0 = \frac{1}{2}[1, 1, 1, 1]^T$
- $\hat{\vec{b}}_1 = \frac{1}{2}[1, j, -1, -j]^T$
- $\hat{\vec{b}}_2 = \frac{1}{2}[1, -1, 1, -1]^T$
- $\hat{\vec{b}}_3 = \frac{1}{2}[1, -j, -1, j]^T$

* verify orthonormality

→ 4c gives $\langle \vec{W}_k, \vec{W}_p \rangle = 0$ for $k \neq p$

scaling both by $1/2$ → keeps inner products at 0

$\rightarrow$ norms : $\|\hat{\vec{b}}_k\|^2 = \left\langle \frac{1}{2}\vec{w}_k , \frac{1}{2}\vec{w}_k \right\rangle = \frac{1}{4}\left\langle \vec{w}_k, \vec{w}_k \right\rangle =$

$\frac{1}{4} \cdot 4 = 1$

$\rightarrow \{ \hat{\vec{b}}_0, \hat{\vec{b}}_1, \hat{\vec{b}}_2, \hat{\vec{b}}_3 \}$ is an orthonormal basis for $\mathbb{C}^4$

* matrix view

$$F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

* for any $\vec{x} \in \mathbb{C}^4$

$$\text{coeff}_k = \langle \hat{\vec{b}}_k, \vec{x} \rangle = \langle \tfrac{1}{2}\vec{w}_k, \vec{x} \rangle = \frac{1}{2}\sum_{n=0}^{3} e^{j2\pi kn/4} x[n] =$$

$$\frac{1}{2}\sum_{n=0}^{3} e^{-j2\pi kn/4} x[n]$$

dft def:

$$X[k] = \sum_{n=0}^{3} x[n] e^{-j2\pi kn/4}$$

$\hookrightarrow$ then $coeff_k = \frac{1}{2} X[k] = \frac{1}{2} X[k] = \frac{1}{\sqrt{N}} X[k]$ with $N=4$

* reconstruction:

$$\vec{x} = \sum_{k=0}^{3} coeff_k \; \tilde{\vec{b_k}} = \sum_{k=0}^{3} \left(\frac{1}{2} X[k]\right)\left(\frac{1}{2} \vec{w_k}\right) =$$

$$\frac{1}{4} \sum_{k=0}^{3} X[k] \vec{w}_k$$

* an orthonormal set is obtained by dividing each $\vec{w_k}$ by 2:

$$\hat{\vec{b}}_k = \frac{1}{2}\left[e^{j2\pi kn/4}\right]_{n=0}^{3},$$

$$k=0,1,2,3$$

↳ these 4 vecs are mutually orthogonal & each has a unit norm

e)

* coordinate transformation matrix

$$T_{\hat{e}\to\hat{b}} = \begin{bmatrix} \hat{b}_0^{\dagger} \\ \hat{b}_1^{\dagger} \\ \hat{b}_2^{\dagger} \\ \hat{b}_3^{\dagger} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

* apply it to $\vec{v}$ to get coordinates in the $\hat{b}_k$ basis

given $\vec{v} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{U} = T_{\hat{e} \to \hat{b}} \vec{v}$,

- $V_0 = \frac{1}{2}(0+1+2+3) = 3$
- $V_1 = \frac{1}{2}(0+(-j)\cdot 1+(-1)\cdot 2+j\cdot 3)$
  $= \frac{1}{2}(-2+2j) = \boxed{-1+j}$
- $V_2 = \frac{1}{2}(0+(-1)\cdot 1+1\cdot 2+(-1)\cdot 3) =$
  $\frac{1}{2}(-2) = \boxed{-1}$
- $V_3 = \frac{1}{2}(0+j\cdot 1+(-1)\cdot 2+(-j)\cdot 3) =$
  $\frac{1}{2}(-2-2j) = \boxed{-1-j}$

$\rightarrow \vec{V} = \begin{bmatrix} 3 \\ -1+j \\ -1 \\ -1-j \end{bmatrix}$

f) reconstruct $\vec{v}$ from its $\hat{b}_k$-coordinates

* inverse (back to $\hat{e}_i$)

$U = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$, so $\vec{v} = U\vec{V}$

$\cancel{x}$ $\quad \vec{v}' = U\vec{v}$

- first entry:

$$v[0] = \frac{1}{2}\left(3 + (-1+j) + (-1) + (-1-j)\right) =$$

$$\frac{1}{2}(0) = \boxed{0}$$

- Second:

$$v[1] = \frac{1}{2}\left(3 + (-1+j)j + (-1)(-1) + (-1-j)(-j)\right) = \frac{1}{2}\left(3 + (-j-1) + 1 + (j-1)\right) =$$

$$\frac{1}{2}(2) = \boxed{1}$$

- third: $\frac{1}{2}\left(3 + (1-j) - 1 + (1+j)\right) =$

$$\frac{1}{2}(4) = \boxed{2}$$

- fourth:

$$v[3] = \frac{1}{2}\left(3 + (\,)+1\right) + 1(-)+1)) =$$

$$\frac{1}{2}(6) = \boxed{3}$$

$$\vec{v} = U\hat{v} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

g) Transform from $\{\hat{b}_k\}$ to $\{\hat{e}_i\}$ and apply to $\vec{v}$

* matrix

$$T_{\hat{e} \to \hat{b}} = U^{\dagger} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$\vec{V} = \begin{bmatrix} 3 \\ -1+j \\ -1 \\ -1-j \end{bmatrix}$$

$$\vec{v} = T_{\hat{b} \to \hat{e}} \vec{V} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 3 \\ -1+j \\ -1 \\ -1-j \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

5)

a) forward transform matrix

for $N = 8$ DFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N},$$

$N = 8, \quad k = 0, \dots, 7$

$$F_8 = \begin{bmatrix} W^{0\cdot0} & W^{0\cdot1} & W^{0\cdot2} & W^{0\cdot3} & W^{0\cdot4} & W^{0\cdot5} & W^{0\cdot6} & W^{0\cdot7} \\ W^{1\cdot0} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & W^{1\cdot7} \\ W^{2\cdot0} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & W^{2\cdot7} \\ W^{3\cdot0} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & W^{3\cdot7} \\ W^{4\cdot0} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & W^{4\cdot7} \\ W^{5\cdot0} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & W^{5\cdot7} \\ W^{6\cdot0} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & W^{6\cdot7} \\ W^{7\cdot0} & W^{7\cdot1} & W^{7\cdot2} & W^{7\cdot3} & W^{7\cdot4} & W^{7\cdot5} & W^{7\cdot6} & W^{7\cdot7} \end{bmatrix}, \quad W = e^{-j\pi/4}$$

* hopefully this illustrates the pattern

b) $N = 8$ DFT

$$x = [1, 0, 0, 0, 0, 0, 0, 0]^T$$

$$X[k] = \sum_{n=0}^{7} x[n] e^{-j2\pi kn/8} = 1 \cdot e^{-j2\pi k \cdot 0/8} =$$

$$\boxed{1} \Rightarrow X = [1,1,1,1,1,1,1,1]^T$$

A impulse at $n=0$ transforms to flat spectrum

Geometric sum of roots of unity

$$X[k] = \sum_{n=0}^{7} e^{-j2\pi kn/8} = \begin{cases} 8, & k=0, \\ 0, & k=1,\dots,7, \end{cases}$$

$$X = [8,0,0,0,0,0,0,0]^T$$

↳ all energy at $k=0$ (DC sequence)

$x = [2, 1, 0, 0, 0, 0, 0, 1]^T$

$n = 0, 1, 7$ are non-zero

$X[k] = 2 \cdot e^{-j2\pi k \cdot 0/8} + e^{-j2\pi k \cdot 1/8} + 1 \cdot e^{-j2\pi k \cdot 7/8} =$

$2 + e^{-j\theta k} + e^{-j(7\theta k)}$ with $\theta k = \frac{2\pi k}{8} = \boxed{\frac{\pi k}{4}}$

evaluated for $K=0,\dots,7$:

K	0	1	2	3	4	5	6	7
$\cos(\pi K/4)$	1	$\frac{\sqrt{2}}{2}$	0	$-\frac{\sqrt{2}}{2}$	-1	$-\frac{\sqrt{2}}{2}$	0	$\frac{\sqrt{2}}{2}$
$X[K]$	4	$2+2\sqrt{2}$	2	$2-\sqrt{2}$	0	$2-\sqrt{2}$	2	$2+\sqrt{2}$

$$X = [4, 2+\sqrt{2}, 2, 2-\sqrt{2}, 0, 2-\sqrt{2}, 2, 2+\sqrt{2}]^T$$